

—— // REWRITING THE DESKTOP // ——

The GUI Automation Bus — Action Reference

Every scriptable verb of every GUI app

ACTION REFERENCE

every verb · every app · one bus

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GUI Script Actions – Global Catalog

Every scriptable **GUI-Script action** (automation-bus verb) exposed by every MenkeTechnologies GUI app — the exact surface `App::open("<app>")->verbs()` returns over the **GUI Automation Bus**: the LIVE runtime surface (appShell verbs + per-app `opts.commands` + dynamically-registered verbs), read from each running app.

483 actions across **2 apps**. Read from each app's live bus surface by `bin/gen-gui-actions-live` — do not hand-edit. Requires every app open when generated.

App	Verbs	Surface
<code>zpdf</code>	322	Acrobat/Preview-style PDF engine — render, edit, annotate, forms, OCR, redact
<code>zwire</code>	161	Chromium-superset browser — tabs, windows, tab-groups, downloads, reading list, power



Acrobat/Preview-style PDF engine — render, edit, annotate, forms, OCR, redact
322 verbs · live bus surface · call as `App::open("zpdf")->call("<verb>", %args)`

(top-level) (322)

accessibility_check
add_3d
add_background
add_barcode
add_bookmark
add_callout
add_data_matrix
add_goto_link
add_grid_overlay
add_header_footer
add_image
add_ink
add_launch_link
add_line_numbers
add_link
add_markup
add_measure
add_movie
add_named_action_link
add_named_destination
add_note
add_page_numbers
add_printer_marks
add_qr_code
add_remote_goto_link
add_rich_media
add_screen
add_sound
add_submit_button
add_text
add_text_with_font
add_thread
add_typed_signature
adjust_image
adjust_page
apply_actions
apply_redactions
attach_file
attachments
auto_crop_margins
auto_link_urls
auto_outline
auto_tag
bates_number
binarize
booklet_order
build_toc_page
canonical_bytes

certify
clear_metadata
clear_recents
clear_signature
color_separations
compare
contact_sheet
content_fingerprint
convert_to_cmyk
copy_fidelity_audit
create_checkbox
create_choice_field
create_from_images
create_layer
create_ocmd
create_push_button
create_radio_group
create_signature_field
create_text_field
crop_page
decrypt
decrypt_pubkey
delete_annotation
delete_annotations_by_author
delete_annotations_by_type
delete_attachment
delete_bookmark
delete_field
delete_layer
delete_named_destination
delete_pages
deskew_page
detect_image_regions
detect_skew_angle
discard_search_index
discard_thumbnails
doc_js_names
document_id
downsample_images
draw_bezier
draw_line
draw_path
draw_rect
duplicate_pages
edit_text
encrypt
encrypt_aes256
encrypt_aes256_with_permissions
encrypt_pubkey
encrypt_pubkey_multi
erase_ink_at
export_comments_csv
export_comments_fdf
export_comments_xfdf
export_docx
export_fdf
export_html
export_markdown
export_page_svg
export_png

export_pptx
export_xfa
export_xfdf
export_xlsx
extract_attachment
extract_fonts
extract_images
extract_tables
extract_text
extract_text_in_region
extract_to
find_replace_text
flatten_annotations
flatten_form
flatten_layers
flatten_transparency
flip_page
form_fields
generate_thumbnails
has_acroform
has_permissions_dict
has_xfa
hidden_content_audit
highlight_search
image_alt_texts
import_comments_fdf
import_comments_xfdf
import_fdf
import_xfdf
ink_coverage
insert_blank_page
insert_pages
interleave
invert_colors
is_encrypted
is_signed
is_tagged
last_document
linearize
links
list_annotations
list_field_actions
list_fonts
list_inks
list_layers
list_output_intents
make_portfolio
make_searchable
mark_visual_differences
merge_file
move_bookmark
move_field
move_page
move_text
n_up
named_destinations
object_stats
ocr_page
ocr_page_words
open_pdf

optimize
outline
overlay_page
page_boxes
page_dimensions
page_size
page_text_runs
page_visual_difference
pdf_info
permissions
preflight
print_to_pdf
read_threads
readability_stats
reading_order
recalculate_fields
recent_documents
recompress_streams
redact_pii
redact_regex
redact_search
redaction_leak_audit
reflow
regenerate_document_id
remove_blank_pages
remove_doc_js
rename_bookmark
rename_field
rename_layer
render_page
reorder_pages
repair
replace_image
replace_pages
reset_form
resize_all_pages
resize_page
restyle_text
reverse_pages
revision_extract
revision_history
rotate_all
rotate_page
run_javascript
sanitize
save_pdf
scan_pii
search
set_all_annotation_flags
set_all_page_boxes
set_annotation_author
set_annotation_border
set_annotation_color
set_annotation_contents
set_annotation_flags
set_annotation_opacity
set_annotation_rect
set_annotation_subject
set_bookmark_action
set_bookmark_level

set_bookmark_style
set_bookmark_target
set_calculation_order
set_doc_js
set_document_action
set_document_language
set_field
set_field_alignment
set_field_appearance
set_field_calculation_js
set_field_colors
set_field_default_value
set_field_export_name
set_field_flags
set_field_format_js
set_field_keystroke_js
set_field_rich_value
set_field_tooltip
set_field_validate_js
set_image_alt_text
set_info_property
set_layer_locked
set_layer_usage
set_layer_visibility
set_metadata
set_open_action_page
set_outline
set_output_intent
set_page_action
set_page_box
set_page_duration
set_page_labels
set_page_layout
set_page_mode
set_page_transition
set_print_preset
set_reading_direction
set_tab_order
set_text_field_maxlen
set_text_field_options
set_trapped
set_user_unit
set_viewer_preference
set_xfa_datasets
set_xmp_metadata
sign
sign_image
sign_visible
signature_count
similarity_score
space_audit
split_by_bookmarks
split_by_count
split_by_ranges
split_by_size
split_by_text
split_odd_even
split_page_grid
split_scanned_images
stamp_image

structure_diff
stylize_page
suggest_filename
swap_pages
tag_pdf_ua
take_launch_file
to_grayscale
to_pdf_a
to_pdf_x
to_single_page
undo_annot
unembed_fonts
validate_fields
validate_pdf_a
validate_pdf_ua
vcs_bisect
vcs_blame
vcs_blame_lines
vcs_checkout
vcs_commit
vcs_diff
vcs_head
vcs_log
vcs_page_objects
verify_full
verify_redaction
verify_signatures
watermark_image
watermark_text
whiteout
word_diff
xfa_datasets
xfa_packets
xmp_metadata

zwire

Chromium-superset browser — tabs, windows, tab-groups, downloads, reading list, power
161 verbs · live bus surface · call as `App::open("zwire")->call("<verb>", %args)`

(`top-level`) (62)

```
clipboard_get
clipboard_set
exec
fs_append
fs_list
fs_mkdir
fs_read
fs_rm
fs_stat
fs_tail
fs_walk
fs_watch
fs_write
get
hello
hook_fire
hooks_delete
hooks_events
hooks_get_script
hooks_list
hooks_save
hooks_script_path
hooks_set_enabled
hooks_set_script
hooks_test_run
hostinfo
hostlog
job_list
job_poll
job_result
job_start
kill
kv_del
kv_get
kv_keys
kv_merge
kv_set
meter_stream
notify
open
peer
peer_connect
peers
ping
ps
pty_kill
pty_resize
pty_spawn
```

```
pty_write
pub
stryke_lsp_send
stryke_lsp_start
stryke_lsp_stop
stryke_run
sub
sysinfo_once
sysinfo_start
sysinfo_stop
unsub
watch_list
watch_stop
which
```

browser (99)

```
browser.activate
browser.addHistoryUrl
browser.addReadingList
browser.allowSleep
browser.bookmarkFolder
browser.bookmarkTab
browser.cancelDownload
browser.centerWindow
browser.clearAllData
browser.clearCache
browser.clearCacheAndCookies
browser.clearCookies
browser.clearDownloads
browser.clearHistory
browser.clearPasswords
browser.closeDuplicates
browser.closeLeft
browser.closeOthers
browser.closeRight
browser.closeTab
browser.closeWindow
browser.collapseGroups
browser.deleteHistoryUrl
browser.detectLanguage
browser.disableExtension
browser.discardTab
browser.download
browser.duplicateTab
browser.enableExtension
browser.expandGroups
browser.extensionOptions
browser.firstTab
browser.fullscreenWindow
browser.goBack
browser.goForward
browser.gotoTab
browser.groupTabs
browser.home
browser.incognitoWindow
browser.keepAwake
browser.keepDisplayAwake
browser.lastTab
browser.launchApp
browser.maximizeWindow
```

browser.mergeWindows
browser.minimizeWindow
browser.moveTabFirst
browser.moveTabLast
browser.moveTabLeft
browser.moveTabRight
browser.moveWindowNextDisplay
browser.muteAll
browser.muteOthers
browser.muteTab
browser.newTab
browser.newWindow
browser.nextTab
browser.nextWindow
browser.notify
browser.open
browser.openDownload
browser.openTab
browser.pauseDownload
browser.pinAll
browser.pinTab
browser.prevTab
browser.prevWindow
browser.reload
browser.reloadAll
browser.reloadHard
browser.removeBookmark
browser.removeReadingList
browser.reopenTab
browser.restoreWindow
browser.resumeDownload
browser.retryDownload
browser.screenshot
browser.showDownload
browser.showDownloads
browser.snapBottom
browser.snapBottomLeft
browser.snapBottomRight
browser.snapLeft
browser.snapRight
browser.snapTop
browser.snapTopLeft
browser.snapTopRight
browser.sortTabs
browser.tabToNewWindow
browser.tmux
browser.ungroupTabs
browser.uninstallExtension
browser.unmuteAll
browser.unmuteTab
browser.unpinAll
browser.unpinTab
browser.zoomIn
browser.zoomOut
browser.zoomReset